4/13 Cpts 350

HW Hint: (Do all hw problems).

Content of this wk and next wk will take 25% of the final.

p1. (i). $\Box(\text{yellow} \vee \Diamond \text{blue})$

It's always true (on the co-walk that we are looking for) that: the current node is yellow or in the future, there is a blue.

for each current node on the co-walk.

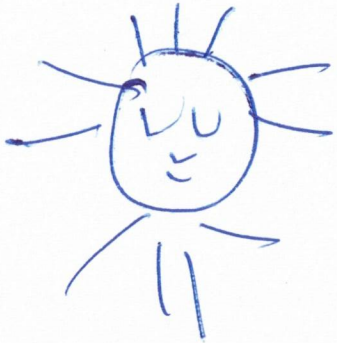
P1. (2). $\Box\Diamond(\text{yellow} \vee \Diamond \text{blue})$

i.o.

There are inf. many times such that, for each such time, it's yellow, or, in the future, there is blue.

Two case: ① inf many blue's

② inf many yellows.



design algs for ① and for ②.



two cases:

(1). we have inf. many blue's on the 2-walk;

(2). we only have finitely many blue's and after the last blue, all yellows are on the 2-walk;



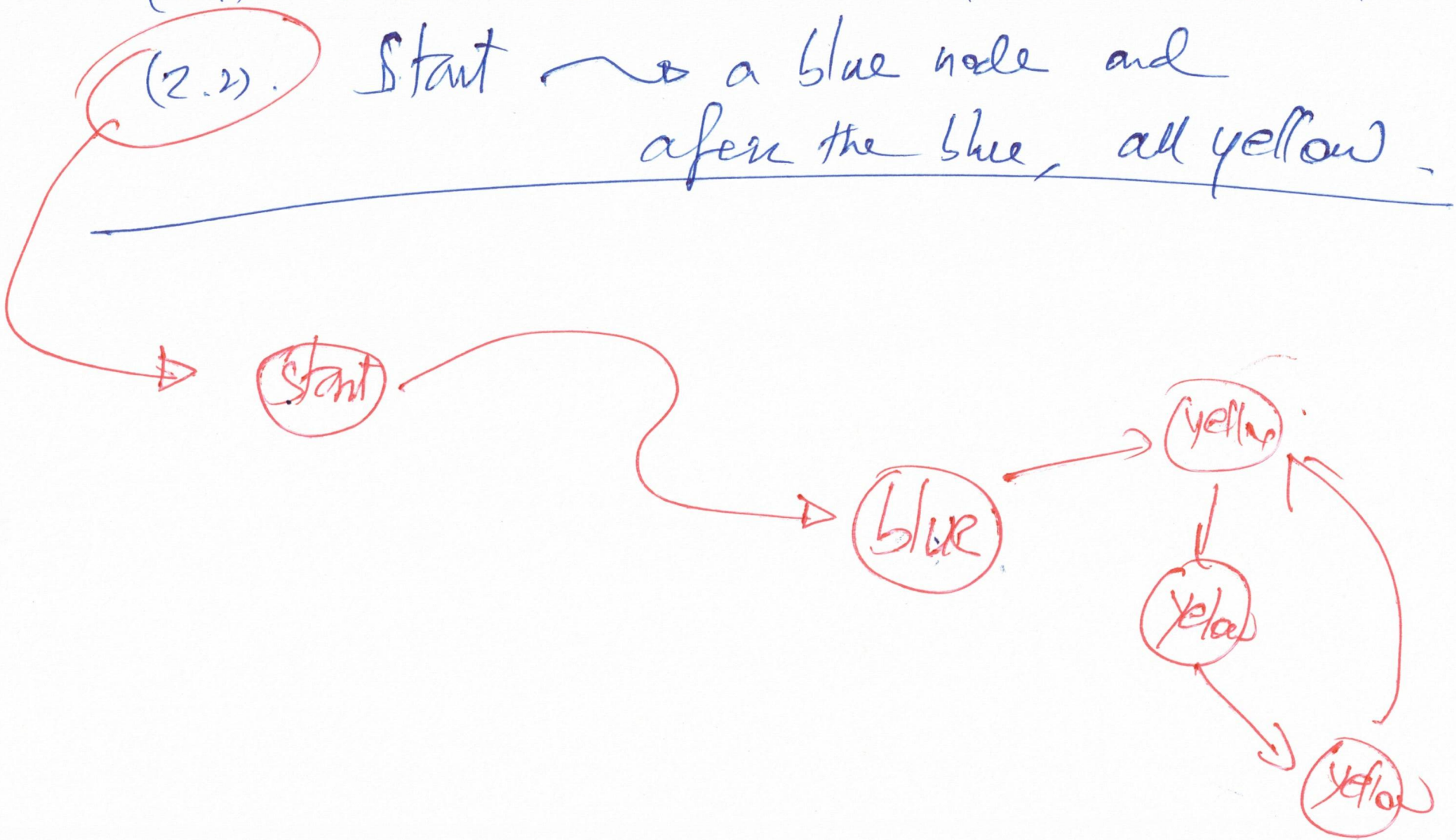
design algs for (1) and for (2).

(2)

For (2).

(2.1). the walk contains all yellow, or

(2.2). Start → a blue node and after the blue, all yellow.



(2.2). From G, I drop all node with non-yellow color. As a result, I have a graph G'.

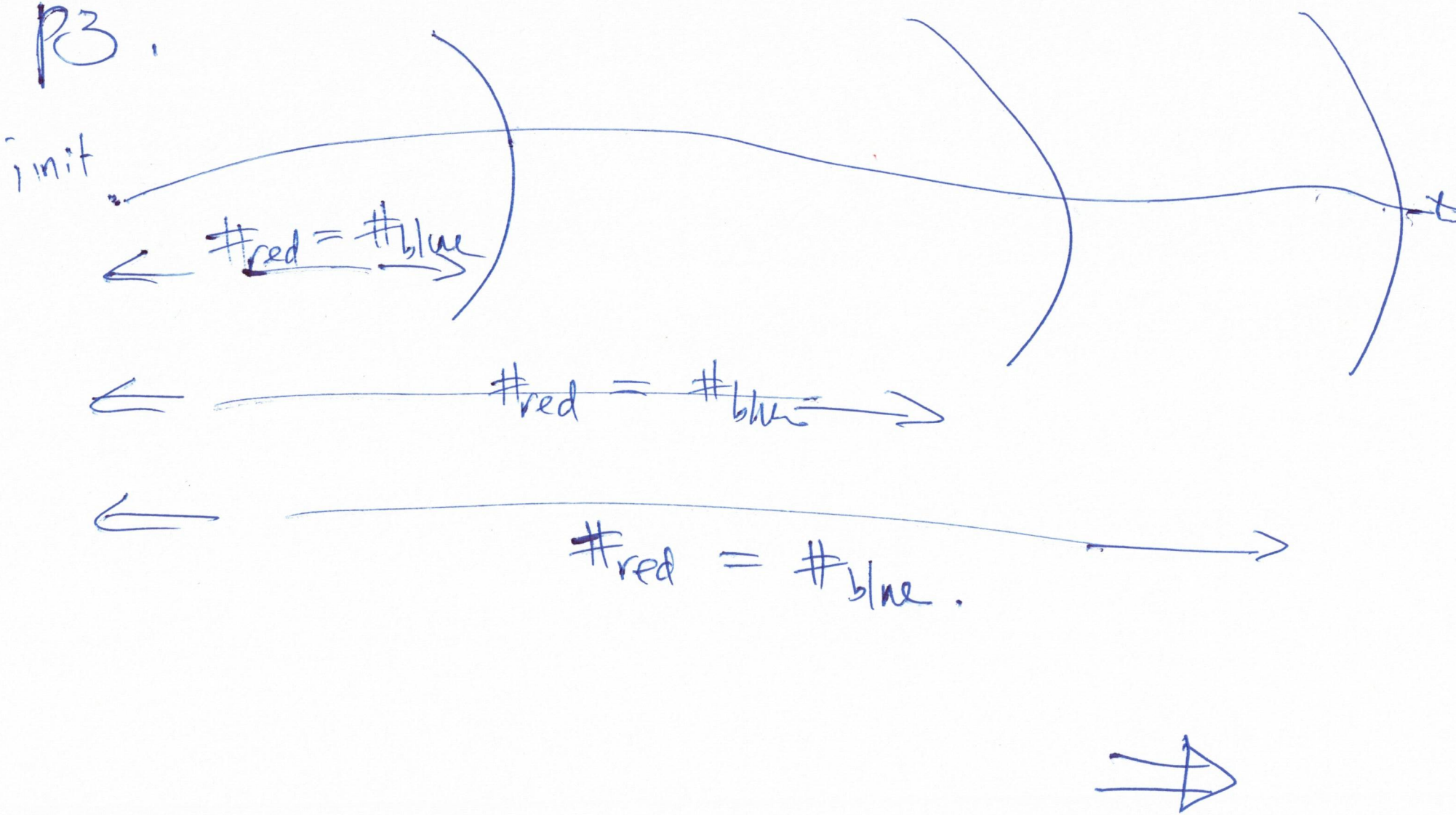
Run SCC on G'. Then I check:

start can reach a blue node (DFS) and the blue has an edge to a yellow node contained in a looping SCC of G'.

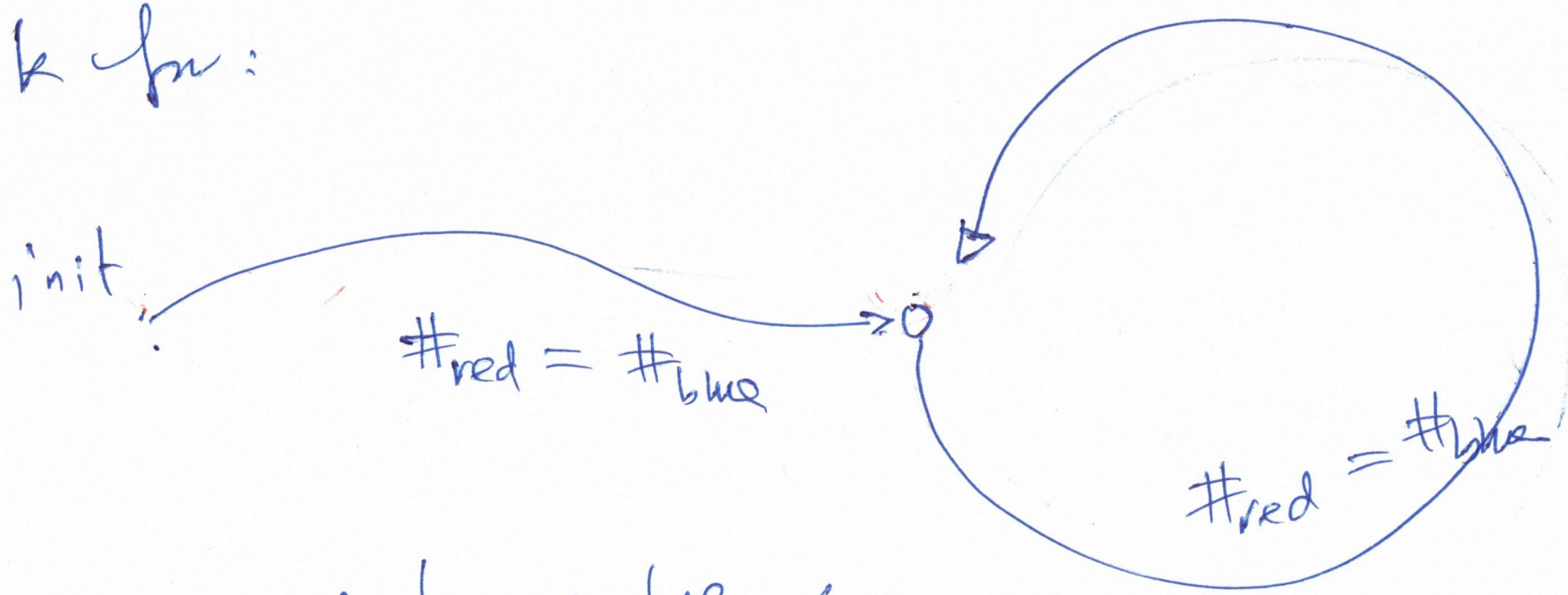
Note that, the above can be done in linear time.

P2. look at class notes.

P3.



look for:



I only need to solve:

(*) Design an alg to decide whether there is a walk $u \rightsquigarrow v$ on which $\#_{red} = \#_{blue}$?

For the G, given two nodes u, v.

I construct a PDA M from the G and u and v:

M is used to simulate a walk from u to v on which M uses the stack to store the difference $|\#_{red} - \#_{blue}|$ on the walk. The sign of $\#_{red} - \#_{blue}$ is stored in M's finite memory. M says yes iff at the end M has empty stack.

Notice that $\binom{*}{*}$ says yes iff $\mathcal{L}(M) \neq \emptyset$.

We know we have alg to solve (317).

P4. (likely in final). However, we only need solve the following:

(***): For the G, for given two nodes u, v, whether there is a walk $u \rightsquigarrow v$ on which $\#_{red} \equiv_5 0$. (you could use SCC to solve it.)

From G, u, v, I construct a FA M to simulate a walk for $u \rightsquigarrow v$ while using nite memory to keep track of the value ($\#_{red}$ mod 5). M says yes iff at the end, the value is zero.

Note that $\begin{pmatrix} * \\ * \\ * \end{pmatrix}$ says yes iff $L(M) \neq \emptyset$.

We do have linear time algs to solve it (3/7).